

JAMES

(JPD AMS DATABASE DESIGN)

1] PREREQUISITE:-

1) Tablenames are in combination of (Train-case) and (snake_case)

e.g.

First letter is always CAPITAL and words are connected with an underscore

e.g. Student_master

2) all columns of the table are written in small letter only.

3) At the end of each table design there is NOTE related to table which describes control of table (C.R.U.D.) and where to add that feature in frontend. Additionally, Primary Key and Foreign Key Reference is mentioned for ease of understanding.

4) Blue colour is **primary key** and **peach colour** is foreign key.

5) Kindly follow all above stated requirements if you are going to add/alter any table or column.

6) It's recommendation to give logical id's wherever required instead of giving random id's.

For e.g. *cs_id* should be in chronological order.

2] TABLE DESIGN:-

1)Api

read_no	rfid_no	read_date	read_time	spid
Integer	Number(2,0)	Date	Time	Varchar2(10)
1 to N	1 to 99	dd/mm/yyyy	hh:mm:ss	2020049910

PK: [read_no](#)

FK: spid => Students

NOTE:

- 1) This table should be cleared at midnight 3:00 AM every day.
- 2) (C.R.U.D) Operation of this table is directly given to faculty on his/her dashboard so that by providing minimal input such as [rfid_no\(classroom_no\),date,time](#) attendance data can be fetched up.
- 3) After that validation of spid can be made using semester checking with spid and for that faculty will

give another **input of semester** for which attendance needed to be taken.

- 4) Don't forget to add **distinct clause** to spid while firing the query so only 1 student attendance can be fetched though he/she has done multiple scans in one lecture.

2)Rfid_uid_spid_map

(Not required if preloaded spid's are there in card)

uid	spid
Varchar2(?)	Varchar2(10)
8 to 20 character long	10 character long

PK : uid

FK : spid => Student

NOTE:

- 1) Read Uid's will be mapped with spid's of students.

3)Users

email	password	user_type
Varchar2(100)	Varchar2(16)	Number(1,0)
harshilramani.mscit20@vsnsngu.ac.in	Alphanumeric password of 16 character long	1 to 9 e.g. student,admin,faculty, Management-staff,etc

PK: email

FK: user_type_id => Users_role

NOTE:

- 1) Admin panel will have all user registering facility and store data to this table as well as login details for students/faculty/admin/management staff(it office) will be taken from this table in order to verify user.
- 2) Email must be of VNSGU only other email is not allowed for signing up.
- 3) C.R.U.D control will be given to Admin.

4)Users_role

user_type	role
Number(1,0)	Varchar2(256)
1,2,3,4....9	System_admin,it-office,faculty,student...

PK: user_type

NOTE:

- 1) Based on user role here dashboard page should be displayed to user on his/her browser.
- 2) C.R.U.D control will be given to Admin panel.

5) Courses

course_id	course_name	total_semester
Number(4,0)	Varchar2(256)	Number(2,0)
9319	M.Sc. IT	1 to 10

PK: (course_name + total_semester)

FK: -

NOTE:

- 1) Used for many purposes like registration, data retrieval, report Etc..
- 2) C.R.U.D. control will be given to admin only.

6) Subjects

subject_id	subject_name	subject_code	semester
Number(4,0)	Varchar2(256)	Number(4,0)	Number(2,0)

16221	OOP,DS,CG...etc	101 to 1010	3
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PK:(subject_name+subject_code)

FK: -

NOTE:

- 1) CRUD control will be given to Admin panel .
- 2) Usage for different purposes.

7) Course_subject_map

cs_id	course_id	subject_id
Number(4,0)	Number(4,0)	Number(4,0)
1 to 9999	1234	6789

PK: cs_id

FK:

- 1)course_id => Courses
- 2)subject_id => Subjects

NOTE:

- 1) NO CRUD over here since its automated although insertion process option must be shown at admin panel in order to ensure correct mapping process.

8)Subject_faculty_year_map

Varchar2(10)	Varchar2(18)	Varchar2(256)	Varchar2(256)	Varchar(256)	Varchar(10)	date	Varchar2(100)	Varchar(15)
Number(4,0)	Year(4)	Number(2,0)	Varchar2(1)	Varchar2(10)				

PK: **spid**

FK:

1)course_id => Course

2)email => Users

NOTE:

- 1) Admin panel has C.R.U.D. control
- 2) Student dashboard data will be fetched from here.
- 3) Don't forget to develop feature of deactivating student
After course completion e.g. status should be
Active/Inactive so report could /*not*/ contain that
unnecessary student data.

10) Faculties

fid	fname	mname	lname	gender	dob	email	contact	joining_year	post
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Varchar2(10)	Varchar2(256)	Varchar2(256)	Varchar2(256)	Varchar(10)	date	Varchar2(100)	Varchar2(15)	Year(4)	Varchar(15)
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PK: [fid](#)

FK: email->Users

NOTE:

- 1) Admin panel has control of CRUD as well as registration details will be stored here.

11)Attendance_master

spid	cs_id	att_date	att_time	att_status
Varchar2(10)	Number(4,0)	date	time	boolean
10 character long	1 to 9999	DD-MM- YYYY	HH:MM:SS PM	True/False

PK: ([spid+cs_id+att_date](#))

FK:

- 1)cs_id => Course_subject_map

- 2)spid => Students

NOTE:

- 1) This is the most sensitive and significant table of database and will be used frequently by all user to perform varieties of actions.
- 2) CRUD control will be given to Faculty and Admin panel
- 3) Management staff user will only fetch records from it and student will also fetch and view(read only).
- 4) att_time has been removed from the primary key because attendance for 1 subject will be taken once in a day and also after taking attendance faculty interface should not allow for second time in individual day (use PHP COOKIES for that in dashboard) . you will get more information in Ams_setup_map table

12)Ams_setup_students_map

spid	ams_setup_id	att_per
Varchar2(10)	Number(8,0)	Varchar2(4)
2020049910	12345678	1% to 100%

PK:(spid+ams_setup_id)

FK:

1)spid => Students

2)ams_setup_id => Ams_setup_faculties_map

NOTE:

- 1) Each student will enrol only once in each setup so composite primary key is given although student can opt for as many as different setup.
- 2) CRUD Control is given to faculty as faculty is going to create SETUP and he/she will add respective students into their setups.
- 3) att_per is attendance percentage which will be shown to students on their dashboard for each setups/subject of respective semester. Apart from that, on faculty dashboard he/she will sort out students based on their area of evaluation by percentages fetched from this table.

13) Ams_setup_course_subject_map

ams_setup_id	cs_id	year
Number(8,0)	Number(4,0)	Year(4)
12345678	1 to 9999	2022 to 9999

PK:ams_setup_id

FK:cs_id =>Courses

NOTE:

- 1) CRUD control will be given to Admin panel as it is automatic process whenever setup is created by faculty process will map it to corresponding cs_id.

2) As mentioned earlier in Attendance_master, you can extra column here for preventing more than one time attendance in individual day OR you may use PHP COOKIES by expiry of 1 day which will block/masked particular setup and resist updating attendance in same day.

14) Ams_setup_faculties_map

ams_setup_id	fid
Number(8,0)	Varchar2(10)
12345678	FID4113422

PK(fid+ams_setup_id)

FK: fid => Faculties

NOTE:

1) CRUD control operation will be given to Faculty as well to Admin panel because at sometime two faculties are teaching same subject and both will take attendance for same setup/subject so admin panel will add manually map 2 faculty into same setup so both can update attendance.(don't forget to add discuss & resolve issue of twice attendance a day confliction here also)